NIKOLA MILOSAVLJEVIC

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WORK EXPERIENCE

DFINITY Stiftung **Zurich, Switzerland**
Site Reliability Engineer / Software Engineer ***Sep 2022 – Present***

- Designed and implemented a highly scalable, efficient service discovery module forming the backbone of the company's observability platform. It continuously synchronizes with global system state to enumerate machines, integrates directly with Prometheus and VictoriaMetrics and exposes structured data used to dynamically generate downstream observability configurations.
    - **Stack:** Rust, Bazel, Python, Docker
- Designed, built and maintained the company's end-to-end logging infrastructure from the ground up. Led the migration for logs from AWS to an on-premise Kubernetes cluster, reducing costs by ~$100K/month and operated a high-throughput pipeline handling ~10M production logs/hour plus ~20M logs/hour for developer test environments with relaxed SLOs.
    - **Stack:** Rust, Vector.dev, Kibana, Elasticsearch, Kubernetes.
- Designed and implemented multiple features to reduce operational burden for node providers (core to the blockchain network), including widely used direct node swaps and node migrations, which are cutting key rotation time from days to seconds, improving security and increasing operational flexibility.
    - **Stack**: Rust
- Built a network topology optimization tool using linear optimization to generate decentralized network layouts under complex business constraints. The tool supports multiple modes to balance decentralization and operational complexity. It can model capacity requirements by estimating the number of additional nodes needed to reach target topologies.
    - **Stack**: Python (pandas, pulp, matplotlib, ...)
- Develop core components on the company's cloud engine platform, including a Web2 backend control panel that enables users to provision and manage sovereign cloud environments and deploy dApps on them, and a Web3 on-chain module for billing and payments, integrating real-world asset swaps and stablecoin-based transactions.
    - **Stack:** Rust, rqlite, Kubernetes, Cloudflare, Docker
- Built a local testing driver using QEMU and libvirt to spin up production-like environments for the cloud engine platform, enabling engineers to run tests against a 1:1 copy of the production instance, significantly shortening feedback loops, reducing reliance on simulations and improving code quality by shifting bug detection earlier in the development cycle.
    - **Stack**: Rust, QEMU, libvirt, Docker
- Designed and implemented an automated release qualification pipeline to evaluate deployment readiness of new candidates. The pipeline provides a production-like test environment, executed performance-intensive workloads pre- and post-upgrade, collects metrics and quantifies regressions or improvements to support data-driven release decisions.
    - **Stack**: Rust, GitHub Actions, Golang
- Contributed broadly to developer experience and internal tooling at DFINITY, including managing testnet infrastructure (testnet farm), improving build systems, developing on-call scheduling and enrollment utilities, and streamlining the planning and rollout of releases, features and hotfixes.

Sotex Solutions (outsourced to Schneider Electric Sevilla) **Novi Sad, Serbia**
Fullstack Software Engineer ***May 2021 – Sep 2022***

- Built a billing determination module used by municipalities to generate utility bills (electricity and water) across regions including Kuwait, Sevilla and the Netherlands. The system integrated with IoT smart meters ingesting terabytes of data. The system was designed for reliable storage and large-scale processing.
    - **Stack**: .NET, Mssql, NServiceBus, Azure, RabbitMQ
- Developed smart meter orchestration systems enabling users and technicians to remotely configure, monitor and diagnose devices using industry-standard protocols, improving observability and operation control at scale.
    - **Stack**: .NET, Mssql
- Refactored project delivery workflows by containerizing applications and implementing Infrastructure-as-code for automated environment provisioning, improving deployment consistency, scalability and release speed.
    - **Stack**: Docker, Ansible, Terraform, AWS, Azure

Sotex Box **Novi Sad, Serbia**
Founding Engineer ***May 2024 – Sep 2025***

- Built and scaled a low-bandwidth ad delivery platform (hobby project turned business) targeting constrained networks. The solution was engineered to run on minimal infrastructure ($40 per device, ~$80/month on operations) while supporting ~10K concurrent devices, currently generating ~$15K/month in revenue.
    - **Stack:** .NET, PostgreSQL, Docker, Pulumi, AWS

EDUCATION

Faculty of Technical Sciences **Novi Sad, Serbia**
Bachelor's and Master's Degrees ***Sept 2018 – Dec 2023***

- Bachelor's GPA: 9.49/10, Master's GPA: 9.75/10
- Worked as a teaching assistant during the last year of Bachelor's and Master's. Was involved with cyber/security and microservice architecture.

ADDITIONAL INFORMATION

Languages:	Advanced English (C1 Cambridge degree), Native Serbian.
Programming languages:	Rust, Python, Golang, .NET (willing to learn more)
Cloud & Infrastructure:	AWS, GCP, Azure, Kubernetes, Docker
CI/CD & Automation:	GitHub Actions, Azure DevOps, Apache Airflow, Bazel
AI:	Heavily utilizing Opencode with Gemini and Claude Opus
Interests:	Gym, Soccer, Tennis, Leetcode, NeoVim, AI, Opencode
Work authorization:	Authorized to work in Switzerland (B Permit) and Serbia, open to relocation

LINKS AND PUBLIC WORK

LinkedIn:	https://www.linkedin.com/in/nikolamilosa20/
GitHub:	https://github.com/NikolaMilosa
Public work:	https://github.com/dfinity/dre, https://github.com/dfinity/ic, https://github.com/dfinity/dre-airflow, https://github.com/dfinity/decentralization